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PEM

2022 Trial Higher School Certificate Examination

Mathematics Advanced

General Instructions

- Reading time – 10 minutes
- Working time – 3 hours
- Write using black pen
- Calculators approved by NESA may be used
- A reference sheet is provided at the back of this paper
- For questions in Section II, show relevant mathematical reasoning
- and/or calculations

Total marks: 100

Section I – 10 marks (pages 2-8)

- Attempt Questions 1-10
- Allow about 15 minutes for this section

Section II – 90 marks (pages 10-32)

- Attempt Questions 11-35
- Allow about 2 hours and 45 minutes for this section

Directions to School or College

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Section I

10 marks

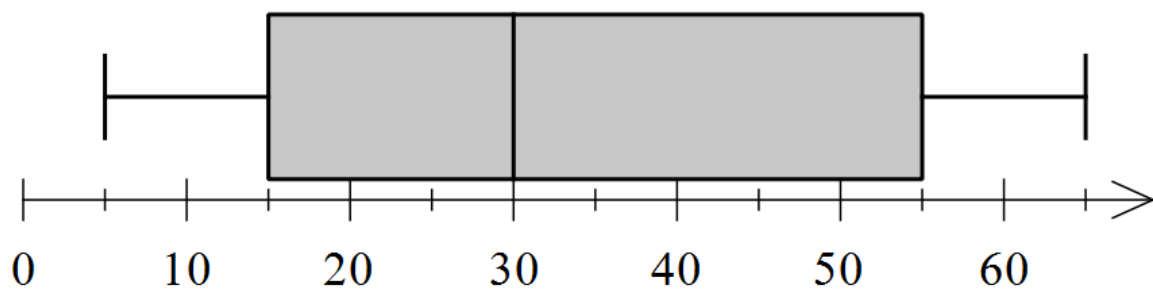
Attempt Questions 1-10

Allow about 15 minutes for this section

Use the multiple-choice answer sheet for Questions 1-10

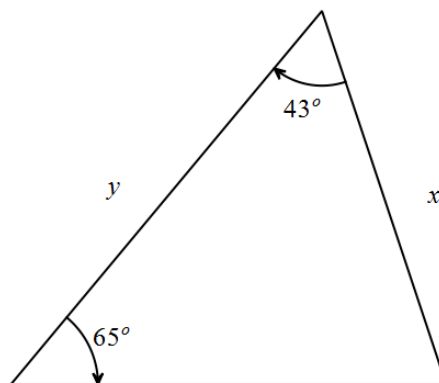
1. The boxplot shows the distribution of online hours spent by a class of students in a certain week. What is the lower quartile of this distribution?

Number of online hours



- A. 5
B. 15
C. 30
D. 55

2. From the diagram, what is the value of y ?



NOT TO SCALE

- A. $y = \frac{x \sin 72^\circ}{\sin 65^\circ}$
- B. $y = \frac{x \sin 43^\circ}{\sin 65^\circ}$
- C. $y = \frac{x \sin 65^\circ}{\sin 72^\circ}$
- D. $y = \frac{x \sin 72^\circ}{\sin 43^\circ}$

3. What is $25 - (2x - 3y)^2$ in fully factorised form?

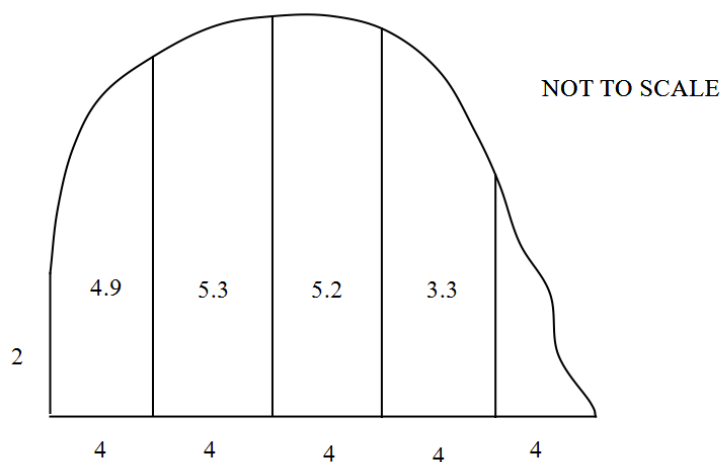
- A. $(5 - 2x - 3y)(5 + 2x + 3y)$
- B. $(5 - 2x - 3y)(5 + 2x - 3y)$
- C. $(5 - 2x + 3y)(5 + 2x + 3y)$
- D. $(5 - 2x + 3y)(5 + 2x - 3y)$

4. What is the value of μ for the probability distribution table for the discrete random variable shown below?

x	1	2	3
$P(X = x)$	0.45	k	0.20

- A. 0.45
- B. 1.00
- C. 1.75
- D. 2.75

5. The diagram below shows a native garden. All measurements are in metres.



What is an approximate value for the area of the native garden using the trapezoidal rule with 5 intervals?

- A. $39m^2$
- B. $58m^2$
- C. $72m^2$
- D. $79m^2$

6. Let $f(x) = 2 - x$ and $g(x) = 6 + x$.

What is the domain of the function $h(x) = \log_e \left(\frac{f(-x)}{g(-x)} \right)$?

- A. $(-2, 6)$
- B. $(-6, 2)$
- C. $(-\infty, -2) \cup (6, \infty)$
- D. $(-\infty, -6) \cup (2, \infty)$

7. A series is given as $1 - 3k + 9k^2 + \dots$, what is the value of k if the sum to infinity is $\frac{3}{4}$?

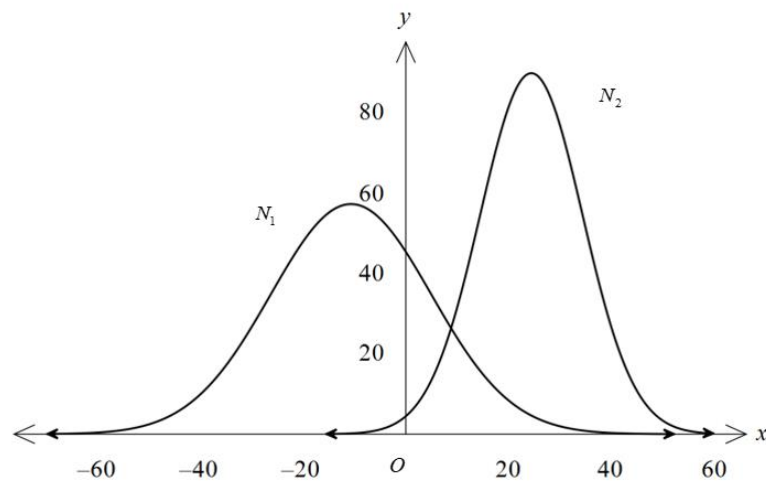
A. $k = -\frac{7}{9}$

B. $k = -\frac{1}{9}$

C. $k = \frac{1}{9}$

D. $k = \frac{7}{9}$

8. Consider the following graphs of two normal distributions, N_1 and N_2 .



Assuming μ_1, σ_1 are the mean and standard deviation of N_1 , while μ_2, σ_2 , are the mean and standard deviation of N_2 , which of the following statements is true?

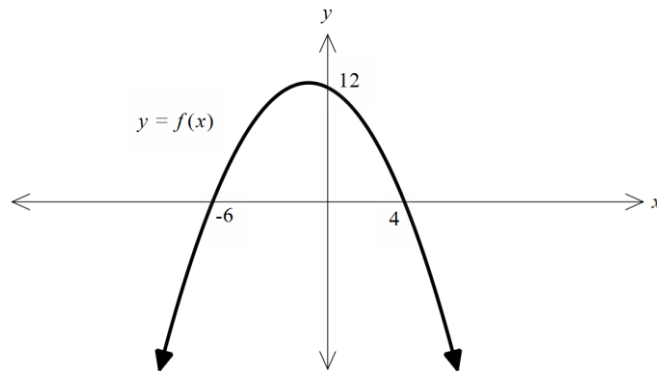
A. $\mu_1 > \mu_2$ and $\sigma_1 > \sigma_2$

B. $\mu_1 > \mu_2$ and $\sigma_1 < \sigma_2$

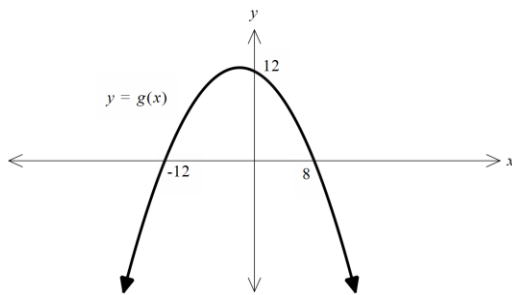
C. $\mu_1 < \mu_2$ and $\sigma_1 > \sigma_2$

D. $\mu_1 < \mu_2$ and $\sigma_1 < \sigma_2$

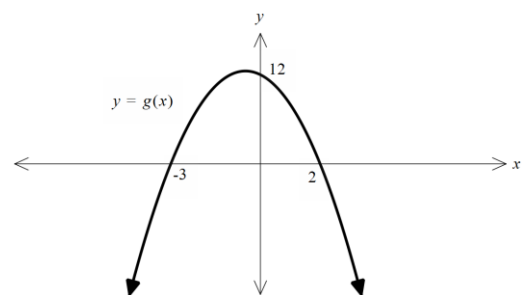
9. The figure below shows the graph of $y = f(x)$. If $g(x) = f\left(\frac{x}{2}\right)$, which of the following may represent the graph of $y = g(x)$?



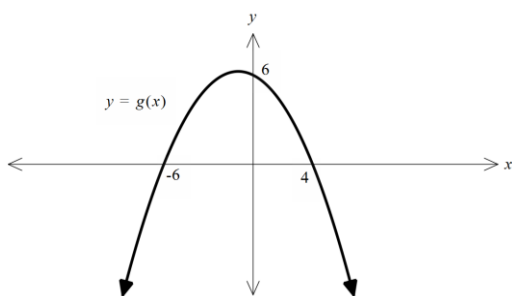
A.



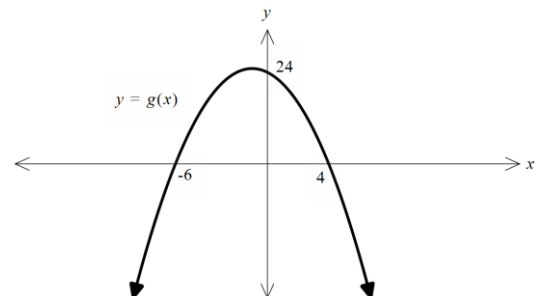
B.



C.

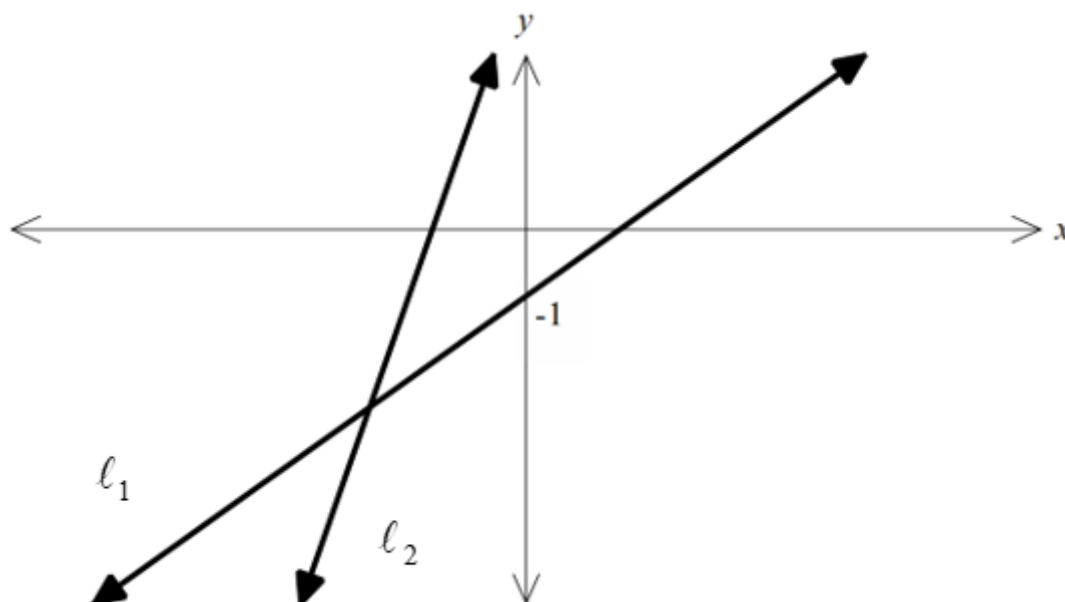


D.



10. The straight lines shown below ℓ_1 and ℓ_2 have equations $x + ay = b$ and $x + cy = d$ respectively. Which of the following are true?

- I. $a < c$
- II. $b > d$
- III. $a + b < c + d$



- A. I and II
- B. I and III
- C. II and III
- D. I, II and III

Trial Higher School Certificate Examination

Mathematics Advanced

Section II Answer Book

90 marks

Attempt Questions 11-35

Allow about 2 hours and 45 minutes for this section

Instructions

- Answer the questions in the space provided. These spaces provide guidance for the expected length of response.
- Your response should include relevant mathematical reasoning and/or calculations.
- Extra writing space is provided at the back of this booklet.
If you use this space, clearly indicate which question you are answering.

Please turn over

Question 11 (2 marks)

Make b the subject of the formula $a(b+5) = 2a + b$.

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Question 12 (2 marks)

Differentiate $\frac{2x}{3x^2 + 1}$.

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Question 13 (4 marks)

Find the following

(a) $\int \sqrt[3]{4x+7} \, dx$. 2

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(b) $\int_0^{\frac{\pi}{6}} \frac{dx}{\cos^2 x}$. 2

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Question 14 (3 marks)

The following table gives the monthly repayment for each \$1 000 borrowed

Term of Loan(years)	6.00%	6.25%	6.50%	6.75%	7.00%	7.25%	7.50%
5	\$19.33	\$19.45	\$19.57	\$19.68	\$19.80	\$19.92	\$20.04
10	\$11.10	\$11.23	\$11.35	\$11.48	\$11.61	\$11.74	\$11.87
15	\$8.44	\$8.57	\$8.71	\$8.85	\$8.99	\$9.13	\$9.27
20	\$7.16	\$7.31	\$7.46	\$7.60	\$7.75	\$7.90	\$8.06
25	\$6.44	\$6.60	\$6.75	\$6.91	\$7.07	\$7.23	\$7.39

- (a) Derek decides to travel during his gap year. He borrows \$20 000 at 6.75% p.a. **1**
from the bank. If he will repay the loan over 15 years, what is Derek's monthly
repayment?

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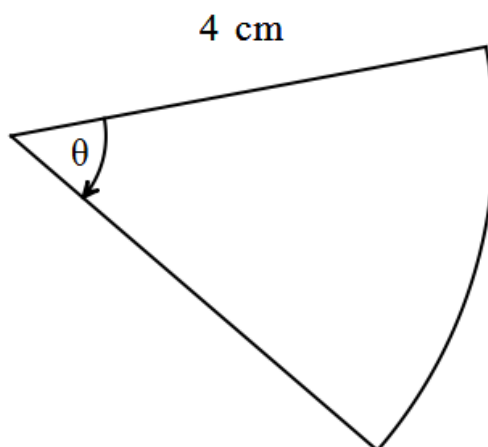
- (b) How much interest does Derek pay on the loan? **2**

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Question 15 (2 marks)

The radius and the area of the sector below are 4 cm and $3\pi \text{ cm}^2$ respectively.

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Calculate the exact perimeter of the sector in terms of π .

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Question 16 (2 marks)

Given that $\log_m p = 1.75$ and $\log_m q = 2.25$, find an expression for $\sqrt[5]{pq^2}$ leaving your answer in terms of m . 2

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Question 17 (2 marks)

Consider the function $f(x) = \ln(x)$. The function $g(x) = af(x) + b$ is a vertical dilation of f by a scale factor of a followed by a vertical translation of f by b units. 2

Express the function $g(x) = \ln(3\sqrt{x})$ in the form of $af(x) + b$, stating the values of a and b .

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Question 18 (3 marks)

The amount of ink in printer cartridges produced by a machine is assumed to be normally distributed with an average of 385 grams of ink and a standard deviation of 6 grams.

- (a) What is the z-score of a cartridge of 397 grams of ink? 1

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- (b) An ink cartridge will be rejected if the amount of ink is more than 2 standard deviations from the average. 2

Determine the probability that an ink cartridge will be rejected.

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Question 19 (5 marks)

A manufacturer stores drums of chemicals. During storage, evaporation takes place. A random sample of 10 drums was taken and the time in storage, x weeks, and the evaporation loss, y ml, are shown in the table below

$x = \text{weeks}$	4	6	7	9	11	13	14	16	17	19
$y = \text{evaporation loss (ml)}$	35	49	52	60	68	78	81	89	87	95

- (a) Find the correlation coefficient r and comment on the strength of the relationship. 2

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- (b) Find the equation of the least squares' regression line. 1

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- (c) Using your model, predict the amount of evaporation that would take place after 35 weeks. 1

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- (d) Comment on the reliability of this prediction from your answer in part (iii). 1

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Question 20 (3 marks)

The seats in a theatre are numbered in numerical order from the first row to the last row. The first row has 14 seats. Each successive row has 4 more seats than the previous one.

- (a) How many seats are there in the fourth row? 1

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- (b) The theatre has a maximum capacity of 1320 seats, what is the greatest number of rows of seats in the theatre? 2

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Question 21 (4 marks)

The frequency distribution table and the cumulative frequency distribution table below show the distribution of the heights of trees in a garden

Height(m)	Frequency
0.1-0.4	a
0.5-0.8	5
0.9-1.2	b
1.3-1.6	12
1.7-2.0	c
2.1-2.4	3

Height less than	Cumulative frequency
0.45	3
0.85	x
1.25	16
1.65	y
2.05	41
2.45	z

- (a) Find the values of x , y and z .

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- (b) Calculate the modal group and median.

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- (c) If a tree is randomly selected, find the probability the height is less than 2.05 but not less than 1.25

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Question 22 (4 marks)

Let $g(x) = \begin{cases} -|x+1| & x < 0 \\ x^2 + 1 & 0 \leq x \leq 3 \end{cases}$

- (a) Sketch $g(x)$ showing all essential features.

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- (b) State the range of $g(x)$.

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Question 23 (3 marks)

The share price (\$P) of a renewable energy company is increasing such that

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$$\frac{dP}{dt} = \frac{3}{t+1},$$

where t is the time in months.

If the initial price of the share price is \$2.00, find the price after 6 months.

Write your answer to the nearest cent.

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Question 24 (4 marks)

The table below show some values of two functions f and g , and of their derivatives f' and g' .

x	1	2	3	4
$f(x)$	4	6	-4	3
$g(x)$	2	0	-1	-5
$f'(x)$	4	8	3	7
$g'(x)$	-5	-4	-6	4

Calculate the following

(a) $\frac{d}{dx}(f(x)g(x))$, when $x = 4$. 2

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(b) $\int_1^3 (g'(x) + 5) dx$. 2

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Question 25 (5 marks)

Two six-sided dice are marked with numbers 1, 1, 2, 2, 3, 5 and 1, 3, 4, 5, 5, 5 respectively. The random variable X is defined to be the minimum of the two numbers obtained when the dice are thrown.

- (a) Fill in the table below giving the probability distribution of X . 2

x	1	2	3	4	5
$P(X = x)$	$\frac{4}{9}$				

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- (b) Find $P(X > 1)$. 1

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- (c) Find $\text{Var}(X)$. 2

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Question 26 (3 marks)

Sketch $f(x) = -(x-1)(x-3)^2(x+1)$ showing all intercepts and necessary features. **3**

You do not need to find any stationary points.

Question 27 (4 marks)

Let X and Y be two events such that $P(X) = 0.5$, $P(Y) = 0.6$ and $P(Y | X) = 0.7$.

(a) Determine whether X and Y are independent events. **3**

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(b) Find $P(X \cup Y)$. **1**

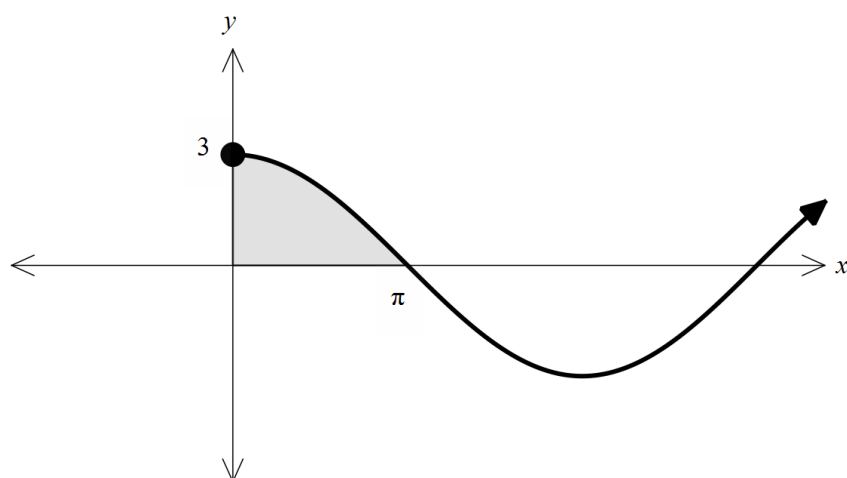
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Question 28 (5 marks)

The graph below represents the function $f(x) = p \cos(qx)$.



- (a) What type of function is $f(x)$? (One-to-one, many-to-one, one-to-many or many-to-many) 1

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- (b) Find the value of p and q . 2

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- (c) Hence, evaluate the area of the shaded region. 2

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Question 29 (4 marks)

John's grandfather wishes to leave him \$250,000 in his will. He proposes to make a payment to an insurance company at the beginning of each year up until his death. The interest rate is set at 7% p.a. At the end of n years after starting his payments, the amount in the fund is A_n . Let his first payment be $\$R$.

- (a) Show that $A_2 = 1.07^2 R + 1.07R$. 1

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- (b) Assuming John's grandfather happens to die in exactly 30 years, immediately after making his last payment. 2

Show that $A_{30} = R \left[\frac{1.07^{31} - 1}{0.07} \right]$.

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- (c) Hence, calculate the annual payment. 1

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Question 30 (3 marks)

Solve the following simultaneous equation.

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$$\log_8 y = x - 2$$

$$3(\log_8 y)^2 = 5 - 2x$$

[illegible]

Question 31 (6 marks)

Let $f(x) = x^3 + x^2 + 3x + 3$.

- (a) Show that the graph of $y = f(x)$ has no stationary points. 2

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- (b) Find any points of inflexion. 2

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- (c) Sketch the graph $y = f(x)$, labelling any points of inflexion and all intercepts with the axes. 2

Question 32 (3 marks)

A continuous random variable X has a probability density function f given

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$$f(x) = \begin{cases} Ax + B & 1 \leq x \leq 7 \\ 0 & \text{otherwise} \end{cases}$$

where A and B are constants. The median is 3. Find the values of A and B .

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Question 33 (3 marks)

Evaluate $\lim_{x \rightarrow 4} \frac{x^2 - 2x - 8}{\sqrt{x} - 2}$.

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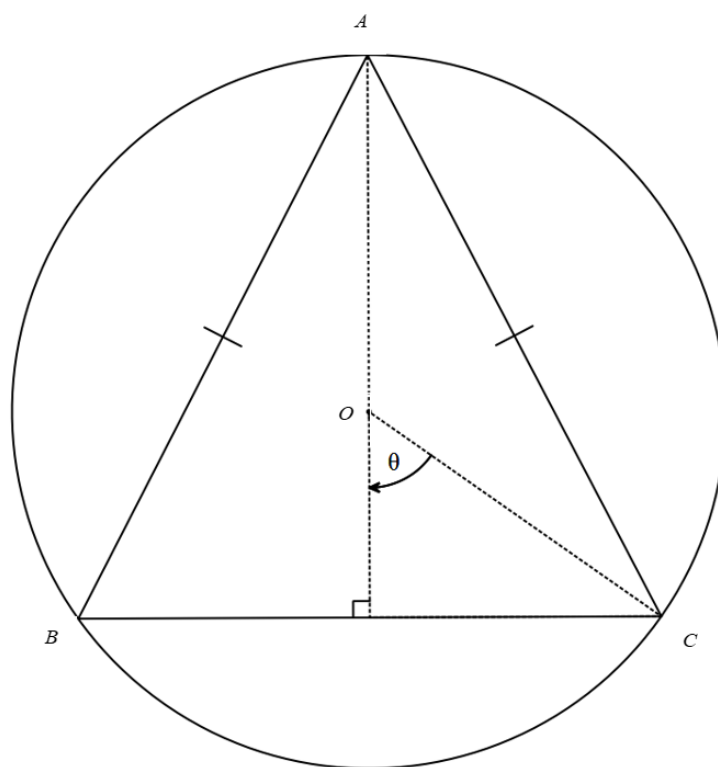
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Question 34 (6 marks)

An isosceles triangle $\triangle ABC$ is inscribed inside a circle of fixed radius 2 and centre O . Let θ be defined as in the diagram below.



- (a) Show that the area of the triangle $\triangle ABC$ is given by $A = 4\sin\theta(1 + \cos\theta)$. 2

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Question 34 continues on page 31

Question 34 (continued)

(b) Find the value of θ that maximises the area of $\triangle ABC$.

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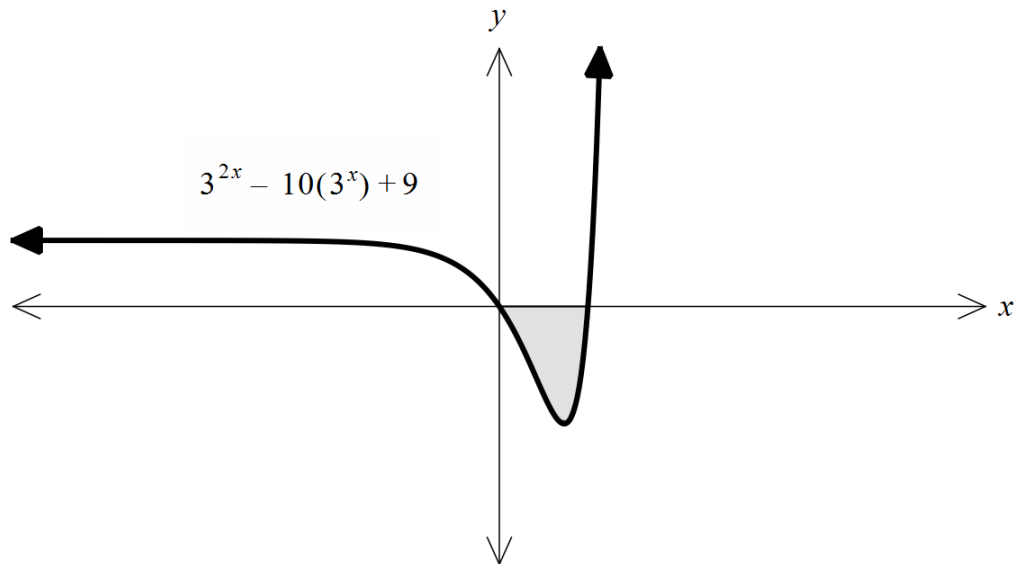
[illegible]

End of Question 34

Question 35 (5 marks)

The graph of $f(x) = 3^{2x} - 10(3^x) + 9$ is shown below. Find the exact value of the area shaded.

5

This image shows a full page of white paper with horizontal dotted lines. The lines are evenly spaced and run across the width of the page, providing a guide for handwriting practice. There are no margins, text, or other markings on the page.

End of paper

Section II extra writing space

If you use this space, clearly indicate which question you are answering.

This image shows a single sheet of white paper with horizontal ruling lines. The lines are evenly spaced and extend across the width of the page. There are no margins, text, or other markings on the paper.

Section II extra writing space

If you use this space, clearly indicate which question you are answering.

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2022 Advanced Mathematics
TRIAL EXAMINATION
SUGGESTED SOLUTIONS

Section 1 Multiple Choice Answer Key

Question	Answer
1	B
2	A
3	D
4	C
5	D
6	A
7	C
8	C
9	A
10	A

Section II

Question 11

Criteria	Marks
Provides correct answer	2
Progress towards solution	1

Sample Answer

$$ab + 5a = 2a + b$$

$$ab - b = -3a$$

$$b(a - 1) = -3a$$

$$b = \frac{-3a}{a - 1}$$

Question 12

Criteria	Marks
Provides correct answer	2
Attempts to use product rule or equivalent merit	1

Sample Answer

$$f(x) = \frac{2x}{3x^2 + 1}$$

$$\begin{aligned} \frac{d}{dx} \left(\frac{2x}{3x^2 + 1} \right) &= \frac{(3x^2 + 1)(2) - (2x)(6x)}{(3x^2 + 1)^2} \\ &= \frac{2 - 6x^2}{(3x^2 + 1)^2} \end{aligned}$$

Question 13 (a)

Criteria	Marks
• Provides correct solution	2
• Progress towards solution	1

Sample Answer

$$\begin{aligned}
 & \int \sqrt[3]{4x+7} \, dx \\
 &= \int (4x+7)^{\frac{1}{3}} \, dx \\
 &= \frac{(4x+7)^{\frac{4}{3}}}{\frac{4}{3}} \times \frac{1}{4} + C \\
 &= \frac{3(4x+7)^{\frac{4}{3}}}{16} + C
 \end{aligned}$$

Question 13 (b)

Criteria	Marks
• Provides correct solution	2
• Integrates primitive function correctly	1

Sample Answer

$$\begin{aligned}
 & \int_0^{\frac{\pi}{6}} \frac{dx}{\cos^2 x} \\
 &= \int_0^{\frac{\pi}{6}} \sec^2 x \, dx \\
 &= \tan x \Big|_0^{\frac{\pi}{6}} \\
 &= \tan \frac{\pi}{6} - \tan 0 \\
 &= \frac{1}{\sqrt{3}}
 \end{aligned}$$

Question 14 (a)

Criteria	Marks
Provides correct solution	1

Sample Answer

(from Table) $20 \times \$8.85 = \177 per month

Question 14 (b)

Criteria	Marks
Provides correct solution	2
Calculates total paid or equivalent merit	1

Sample Answer

Total: $\$177 \times 12 \times 15 = \31860

Interest = $\$31860 - \20000

= $\$11860$

Question 15

Criteria	Marks
Provides correct solution	2
Finds the value of θ or equivalent merit	1

Sample Answer

$$\frac{1}{2}(4)^2 \theta = 3\pi$$

$$\theta = \frac{3\pi}{8}$$

$$P = 4 + 4 + 4\left(\frac{3\pi}{8}\right)$$

$$P = 8 + \frac{3\pi}{2} \text{ cm}$$

Question 16

Criteria	Marks
Provides correct solution	2
Uses a correct index law or equivalent merit	1

Sample Answer

$$p = m^{1.75}, q = m^{2.25}$$

$$pq^2 = m^{1.75} \times m^{4.5}$$

$$pq^2 = m^{6.25}$$

$$\sqrt[5]{pq^2} = (m^{6.25})^{\frac{1}{5}}$$

$$\sqrt[5]{pq^2} = m^{1.25}$$

Question 17

Criteria	Marks
Provides correct solution	2
Uses one correct logarithm law or equivalent merit	1

Sample Answer

$$\ln(3\sqrt{x}) = \ln(3) + \frac{1}{2}\ln(x)$$

$$a = \frac{1}{2}$$

$$b = \ln 3$$

Question 18 (a)

Criteria	Marks
Provides correct solution	1

Sample Answer

$$Z = \frac{X - \mu}{\sigma}$$

$$Z = \frac{397 - 385}{6}$$

$$Z = 2$$

Question 18 (b)

Criteria	Marks
Provides correct solution	2
Calculates only 1 side of the graph (2.5%) or the area in between	1

Sample Answer

$$P(|X| > 397) = P(|Z| > 2)$$

$$P(|X| > 397) = P(Z < -2) \& P(Z > 2)$$

$$P(|X| > 397) = 5\% \text{ (from normal distribution graph)}$$

Question 19 (a)

Criteria	Marks
Provides correct solution	2
Calculates r value or states relationship correctly	1

Sample Answer

$$r = 0.9901098193$$

$$r = 0.99(2d.p)$$

$\therefore$ strong positive linear relationship

Question 19 (b)

Criteria	Marks
Provides correct solution	1

Sample Answer

$$y = A + Bx$$

$$A = 24.117338$$

$$B = 3.903677758$$

$$y = 24.12 + 3.90x$$

Question 19 (c)

Criteria	Marks
Provides correct solution	1

Sample Answer

$$y = 24.12 + 3.9(35)$$

$$y = 160.62$$

$$y = 160$$

Question 19 (d)

Criteria	Marks
Provides correct explanation	1

Sample Answer

Although there is a strong relationship between the variables our x value of 35 is far away from our data set. i.e. we are extrapolating and the value is less reliable the farther away from the data set we used in our calculations

Question 20 (a)

Criteria	Marks
Provides correct solution	1

Sample Answer

$$a = 14, d = 4$$

$$T_4 = 14 + (4 - 1)4$$

$$T_4 = 26$$

26 seats in row 4

Question 20 (b)

Criteria	Marks
Provides correct solution	2
Attempts to use formula or equivalent merit	1

Sample Answer

$$1320 = \frac{n}{2}(2a + (n-1)d)$$

$$1320 = \frac{n}{2}(28 + 4n - 4)$$

$$2640 = (24 + 4n)n$$

$$4n^2 + 24n - 2640 = 0$$

$$\text{as } n > 0, n = \frac{-24 + \sqrt{24^2 - 4(4)(-2640)}}{2(4)}$$

$$n \approx 22.86503431...$$

$$\therefore n = 22$$

$\therefore$ maximum 22 rows

Question 21 (a)

Criteria	Marks
Provides correct solution	1

Sample Answer

$$x = 8, y = 28, z = 44$$

Question 21 (b)

Criteria	Marks
Provides correct solution	2
Progress towards solution	1

Sample Answer

$$a = 3, b = 8, c = 13$$

$$\text{Modal group} = 1.7m - 2.0m$$

$$\text{Median} = \left(\frac{44+1}{2} \right) = 1.45m$$

Question 21 (c)

Criteria	Marks
Provides correct solution	1

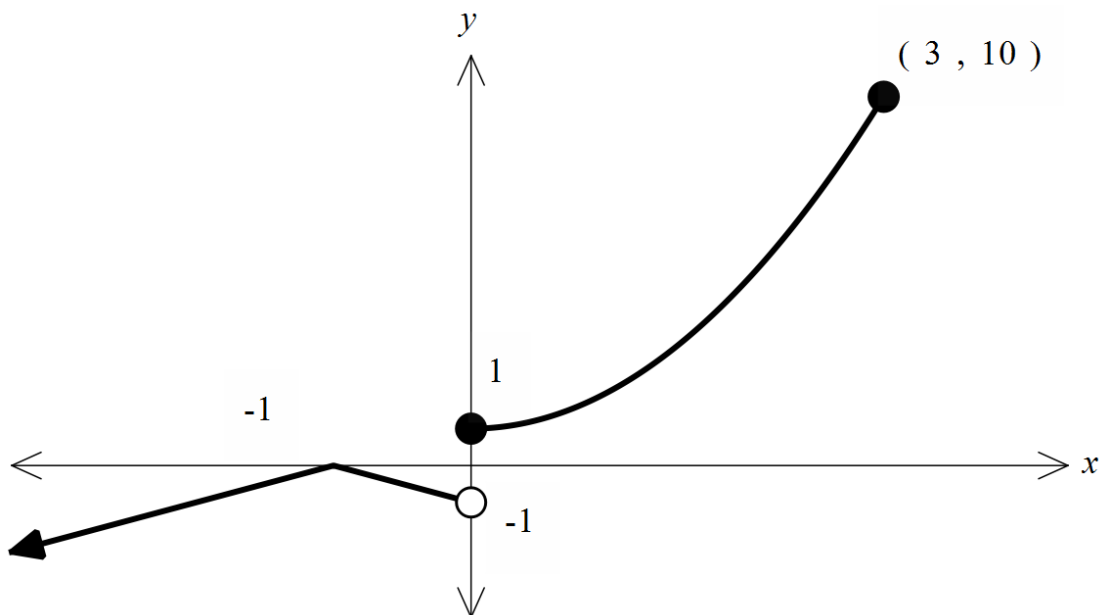
Sample Answer

$$P(1.25 < X < 2.05) = \frac{12+13}{44}$$

$$= \frac{25}{44}$$

Question 22 (a)

Criteria	Marks
Provides correct solution, showing all intercepts with relevant axes and key features such as vertex and endpoint	2
Correct graph shape but missing either intercepts, open/closed circle, or endpoint (3,10)	1

Sample Answer**Question 22 (b)**

Criteria	Marks
Provides correct solution	2
States 1 correct section of range values	1

Sample Answer

$$R: y \leq 0 \text{ or } 1 \leq y \leq 10$$

Question 23

Criteria	Marks
Provides correct solution	3
Finds constant of integration or equivalent merit	2
Correct primitive function or equivalent merit	1

Sample Answer

$$P = 3\ln(t+1) + C$$

$$t = 0, P = 2$$

$$2 = 3\ln(1) + C$$

$$C = 2$$

$$P = 3\ln(t+1) + 2$$

$$P = 3\ln 7 + 2$$

$$P \approx 7.84$$

After 6months, the price is \$7.84

Question 24 (a)

Criteria	Marks
Provides correct solution	2
Attempts to use product rule	1

Sample Answer

$$\begin{aligned}\frac{d}{dx}(f(x)g(x)) &= f'(x)g(x) + f(x)g'(x) \\ f'(4)g(4) + f(4)g'(4) &= 7(-5) + 3(4) \\ &= -23\end{aligned}$$

Question 24 (b)

Criteria	Marks
Provides correct solution	2
Integrates correctly or equivalent merit	1

Sample Answer

$$\begin{aligned}\int_1^3 (g'(x) + 5) dx \\ &= g(x) + 5x \Big|_1^3 \\ &= g(3) + 15 - (g(1) + 5) \\ &= -1 + 15 - 2 - 5 \\ &= 7\end{aligned}$$

Question 25 (a)

Criteria	Marks
• Provides correct solution	2
• One correct value or showing progress towards solution	1

Sample Answer

	1	1	2	2	3	5
1	1	1	1	1	1	1
3	1	1	2	2	3	3
4	1	1	2	2	3	4
5	1	1	2	2	3	5
5	1	1	2	2	3	5
5	1	1	2	2	3	5

x	1	2	3	4	5
$P(X = x)$	$\frac{4}{9}$	$\frac{5}{18}$	$\frac{1}{6}$	$\frac{1}{36}$	$\frac{1}{12}$

Question 25 (b)

Criteria	Marks
• Provides correct solution	1

Sample Answer

$$P(X > 1) = 1 - \frac{4}{9}$$

$$P(X > 1) = \frac{5}{9}$$

Question 25 (c)

Criteria	Marks
Provides correct solution	2
Calculates $E(X)$ correctly or equivalent merit	1

Sample Answer

$$E(X) = 1 \times \frac{4}{9} + 2 \times \frac{5}{18} + 3 \times \frac{1}{6} + 4 \times \frac{1}{36} + 5 \times \frac{1}{12}$$

$$E(X) = \frac{73}{36}$$

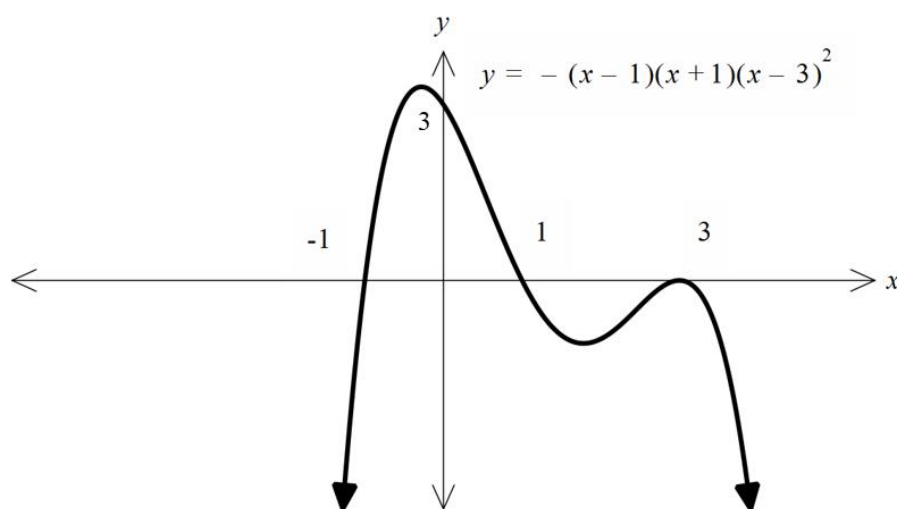
$$E(X^2) = \frac{67}{12}$$

$$\text{Var}(X) = \frac{67}{12} - \left(\frac{73}{36}\right)^2$$

$$\text{Var}(X) = \frac{1907}{1296}$$

Question 26

Criteria	Marks
• Provides correct solution	3
• Correct intercepts or general shape	2
• Calculates one correct intercept or states degree of roots	1

Sample Answer

Question 27 (a)

Criteria	Marks
Provides correct solution, with valid reasoning of independence	3
Finds $P(Y \cap X)$ or $P(X) \times P(Y)$ or equivalent merit	2
Attempts to use conditional probability formula	1

Sample Answer

$$P(Y | X) = \frac{P(Y \cap X)}{P(X)}$$

$$0.7 = \frac{P(Y \cap X)}{0.5}$$

$$P(Y \cap X) = 0.35$$

$$P(Y) \times P(X) = 0.5 \times 0.6$$

$$P(Y) \times P(X) = 0.3$$

$$P(Y) \times P(X) \neq P(Y \cap X)$$

$\therefore$ Not Independent

Question 27 (b)

Criteria	Marks
Provides correct solution	1

Sample Answer

$$P(Y \cup X) = 0.5 + 0.6 - 0.35$$

$$P(Y \cup X) = 0.75$$

Question 28 (a)

Criteria	Marks
Provides correct solution	1

Sample Answer

Many-to-one

Question 28 (b)

Criteria	Marks
Provides correct solution	2
Finds one correct value	1

Sample Answer

$$p = 3, q = \frac{1}{2}$$

Question 28 (c)

Criteria	Marks
Provides correct solution	2
Correct primitive function or equivalent merit	1

Sample Answer

$$A = \int_0^{\pi} 3 \cos\left(\frac{x}{2}\right)$$

$$A = \left[6 \sin\left(\frac{x}{2}\right) \right]_0^{\pi}$$

$$A = 6 \sin \frac{\pi}{2} - 6 \sin 0$$

$$A = 6u^2$$

Question 29 (a)

Criteria	Marks
Shows correct progress	1

Sample Answer

$$A_1 = 1.07 A_0$$

$$A_1 = 1.07 R$$

$$A_2 = (A_1 + R)1.07$$

$$A_2 = 1.07^2 R + 1.07 R$$

Question 29 (b)

Criteria	Marks
Provides correct solution	2
Progress towards solution	1

Sample Answer

$$A_3 = (A_2 + 1.07) R$$

$$A_3 = 1.07^3 R + 1.07^2 R + 1.07 R$$

⋮

$$A_{30} = 1.07^{30} R + \dots + 1.07 R^2 + 1.07 R$$

payout happens after he makes last payment

$$\therefore A_{30} = 1.07^{30} R + \dots + 1.07^2 R + 1.07 R + R$$

$$A_{30} = R \left[1.07^{30} + \dots + 1.07 + 1 \right]$$

GP with $a = 1, r = 1.07, n = 31$

$$\therefore A_{30} = R \left[\frac{1.07^{31} - 1}{0.07} \right]$$

Question 29 (c)

Criteria	Marks
Provides correct solution	1

Sample Answer

$$250000 = R \left[\frac{1.07^{31} - 1}{0.07} \right]$$

$$R = \frac{250000}{\left[\frac{1.07^{31} - 1}{0.07} \right]}$$

$$R = \$2449.226521$$

$$R = \$2449.23$$

Question 30

Criteria	Marks
Provides correct solution	3
Solves quadratic correctly	2
Forms quadratic equation or equivalent merit	1

Sample Answer

$$\log_8 y = x - 2 \quad (1)$$

$$3(\log_8 y)^2 = 5 - 2x \quad (2)$$

Substituting (1) into (2)

$$3(x - 2)^2 = 5 - 2x$$

$$3(x^2 - 4x + 4) = 5 - 2x$$

$$3x^2 - 12x + 12 = 5 - 2x$$

$$3x^2 - 10x + 7 = 0$$

$$(3x - 7)(x - 1) = 0$$

$$x = \frac{7}{3}, x = 1$$

Substituting back into equation (1)

$$\therefore \log_8 y = \frac{7}{3} - 2, \log_8 y = 1 - 2$$

$$\log_8 y = \frac{1}{3}, \log_8 y = -1$$

$$y = \sqrt[3]{8}, y = 8^{-1}$$

$$y = 2, \frac{1}{8}$$

$$\therefore \text{solutions are } \left(\frac{7}{3}, 2\right) \& \left(1, \frac{1}{8}\right)$$

Question 31 (a)

Criteria	Marks
Provides correct solution	2
Differentiates correctly and attempts to solve $f'(x) = 0$ or equivalent merit	1

Sample Answer

Let $f(x) = x^3 + x^2 + 3x + 3$

$$f'(x) = 3x^2 + 2x + 3$$

$$f'(x) = 0$$

$$\therefore 3x^2 + 2x + 3 = 0$$

$$\Delta = (2)^2 - 4(3)(3)$$

$$\Delta = -32$$

$$\Delta < 0$$

$\therefore f'(x) = 0$ has no solution

$\therefore f(x)$ has no stationary points

Question 31 (b)

Criteria	Marks
Provides correct solution	2
Calculates correct angles	1

Sample Answer

$$f''(x) = 6x + 2$$

$$f''(x) = 0$$

$$6x + 2 = 0$$

$$x = -\frac{1}{3}$$

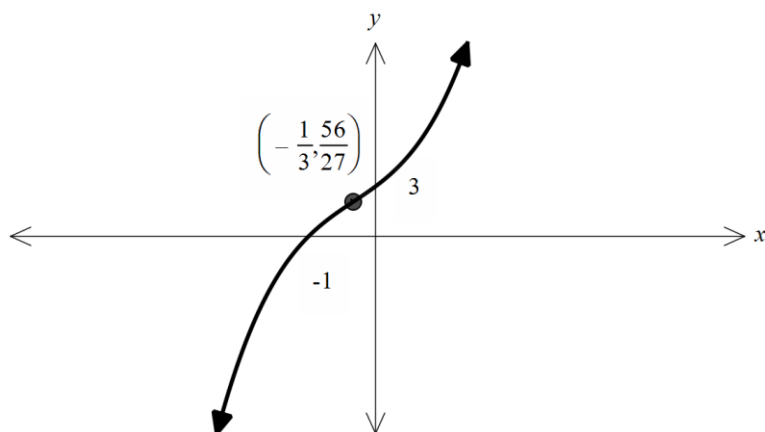
$$f\left(-\frac{1}{3}\right) = \frac{56}{27}$$

x	-1	$-\frac{1}{3}$	0
$f''(x)$	-4	0	2

Since $f''(x)$ changes sign, point of inflexion at $\left(-\frac{1}{3}, \frac{56}{27}\right)$

Question 31 (c)

Criteria	Marks
Draws correct graph with all intercepts and inflexion point	2
Correct shape of graph or missing an essential feature	1

Sample Answer

Question 32

Criteria	Marks
Provides correct solution	3
Creates correct simultaneous equations or equivalent merit	2
Creates one correct integral or equivalent merit	1

Sample Answer

$$\int_1^7 Ax + B = 1 \quad (1)$$

$$\int_1^3 Ax + B = \frac{1}{2} \quad (2)$$

$$\therefore \left[\frac{Ax^2}{2} + Bx \right]_{x=1}^{x=7} = 1 \quad (1)$$

$$\& \left[\frac{Ax^2}{2} + Bx \right]_{x=1}^{x=3} = \frac{1}{2} \quad (2)$$

$$24A + 6B = 1 \dots (1)$$

$$4A + 2B = \frac{1}{2} \dots (2)$$

$$12A + 6B = \frac{3}{2} \dots (3)$$

$$(1) - (3)$$

$$12A = -\frac{1}{2}$$

$$A = -\frac{1}{24}$$

$$B = \frac{1}{3}$$

Question 33

Criteria	Marks
• Provides correct solution	3
• Rationalises denominator and factorises quadratic	2
• Rationalises denominator or factorises quadratic	1

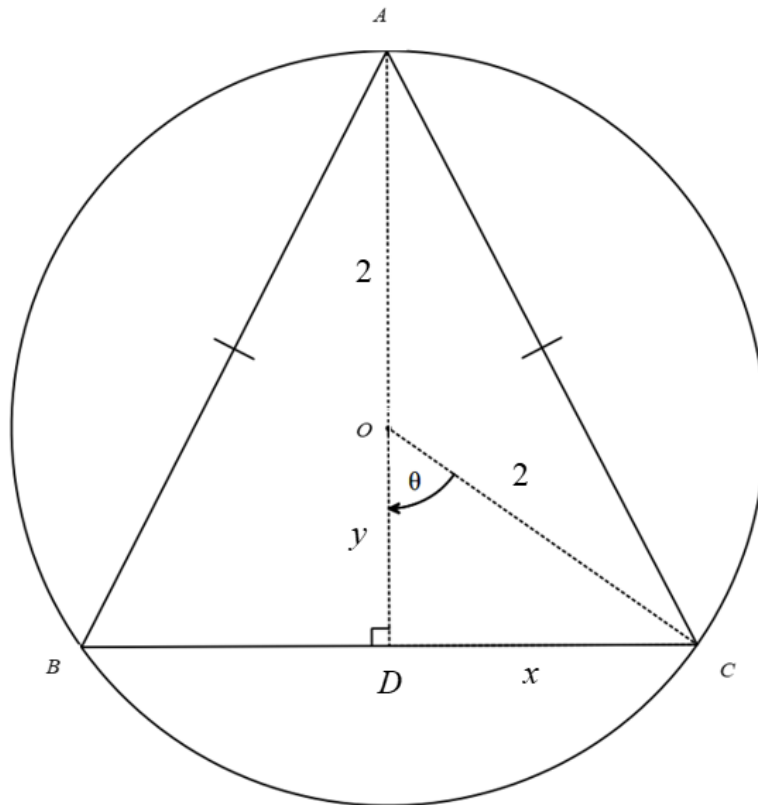
Sample Answer

$$\begin{aligned} & \lim_{x \rightarrow 4} \frac{x^2 - 2x - 8}{\sqrt{x} - 2} \\ &= \lim_{x \rightarrow 4} \frac{(x+2)(x-4)}{\sqrt{x} - 2} \times \frac{\sqrt{x} + 2}{\sqrt{x} + 2} \\ &= \lim_{x \rightarrow 4} \frac{(x+2)(x-4)(\sqrt{x} + 2)}{x - 4} \\ &= \lim_{x \rightarrow 4} (x+2)(\sqrt{x} + 2) \\ &= (4+2)(\sqrt{4} + 2) \\ &= 24 \end{aligned}$$

Question 34 (a)

Criteria	Marks
Provides correct solution	2
Progress towards solution	1

Sample Answer



In $\triangle ODC$,

$$\sin \theta = \frac{x}{2}$$

$$\cos \theta = \frac{y}{2}$$

$$x = 2 \sin \theta$$

$$y = 2 \cos \theta$$

$$\text{Area of } \triangle ABC = \frac{bh}{2}$$

$$b = 4 \sin \theta$$

$$h = 2 + 2 \cos \theta$$

$$A = \frac{8 \sin \theta (1 + \cos \theta)}{2}$$

$$A = 4 \sin \theta (1 + \cos \theta)$$

Question 34 (b)

Criteria	Marks
• Provides correct solution	4
• Finds correct θ values, including testing nature	3
• Attempts to solve $\frac{dA}{d\theta} = 0$, using the identity $\sin^2 \theta + \cos^2 \theta = 1$, or equivalent merit.	2
• Differentiates correctly or equivalent merit	1

Sample Answer

$$A = 4 \sin \theta (1 + \cos \theta)$$

$$\frac{dA}{d\theta} = 4 \cos \theta (1 + \cos \theta) - 4 \sin^2 \theta$$

$$\frac{dA}{d\theta} = 4 \cos \theta + 4 \cos^2 \theta - 4 \sin^2 \theta$$

$$\frac{dA}{d\theta} = 4 \cos \theta + 4 \cos^2 \theta - 4(1 - \cos^2 \theta)$$

$$\frac{dA}{d\theta} = 8 \cos^2 \theta + 4 \cos \theta - 4$$

$$\frac{dA}{d\theta} = 4(2 \cos^2 \theta + \cos \theta - 1)$$

$$\frac{dA}{d\theta} = 0$$

$$\therefore (2 \cos \theta - 1)(\cos \theta + 1) = 0$$

$$\cos \theta = \frac{1}{2}, \theta = \frac{\pi}{3}$$

$$\cos \theta \neq -1, 0 < \theta < \pi$$

x	$\frac{\pi}{6}$	$\frac{\pi}{3}$	$\frac{\pi}{2}$
$\frac{dA}{d\theta}$	> 0	0	< 0
	Increasing	Stationary	Decreasing

$\therefore \theta = \frac{\pi}{3}$ is a maximum S.P.

$$A\left(\frac{\pi}{3}\right) = 4 \sin \frac{\pi}{3} \left(1 + \cos \frac{\pi}{3}\right)$$

$$A = 3\sqrt{3}$$

Question 35

Criteria	Marks
• Provides correct solution in exact form.	5
• Correct substitution and makes progress towards solution	4
• Correctly finds primitive function	3
• Sets up correct integral to calculate area	2
• Finds x – intercepts , or equivalent merit	1

Sample Answer

$$\text{Let } 3^x = u$$

$$0 = u^2 - 10u + 9$$

$$0 = (u - 9)(u - 1)$$

$$u = 1, 9$$

$$\therefore 3^x = 1 \text{ \& } 3^x = 9$$

$$\therefore x = 0, 2$$

As area is below the curve

$$A = \left| \int_0^2 3^{2x} - 10(3^x) + 9 \, dx \right|$$

$$A = \left[\frac{3^{2x}}{2 \ln 3} - \frac{10(3^x)}{\ln 3} + 9x \right]_0^2$$

$$A = \left(\frac{3^4}{2 \ln 3} - \frac{10(3^2)}{\ln 3} + 18 \right) - \left(\frac{3^0}{2 \ln 3} - \frac{10(3^0)}{\ln 3} + 0 \right)$$

$$A = \left(\frac{81 - 180 + 36 \ln 3 - 1 + 20}{2 \ln 3} \right)$$

$$A = \left| \frac{18 \ln 3 - 40}{\ln 3} \right|$$

$$\therefore A = \frac{40 - 18 \ln 3}{\ln 3}$$

2022 PEM Trial HSC Advanced Mathematics

Mapping Grid

Section 1

Question	Marks	Content	Syllabus outcomes	Targeted performance bands
1	1	MA-S2 Descriptive Statistics and Bivariate Data Analysis	MA12-8	2
2	1	MA-T1 Trigonometry and Measure of Angles	MA11-3	2-3
3	1	MA-F1 Working with Functions	MA11-1	2-3
4	1	MA-S1 Probability and Discrete Probability Distributions	MA11-7	3
5	1	MA-C4 Integral Calculus	MA12-3	3-4
6	1	MA-E1 Logarithms and Exponentials	MA11-6	4
7	1	MA-M1 Modelling Financial Situations	MA12-4	4
8	1	MA-S3 Random Variables	MA12-8	4-6
9	1	MA-F2 Graphing Techniques	MA12-1	5-6
10	1	MA-F1 Working with Functions	MA11-1	6

Section II

Question	Marks	Content	Syllabus outcomes	Targeted performance bands
11	2	MA-F1 Working with Functions	MA11-1	2-3
12	2	MA-C1 Introduction to Differentiation	MA11-5	3
13 (a)	2	MA-C4 Integral Calculus	MA12-7	3
13 (b)	2	MA-C4 Integral Calculus	MA12-7	3-4
14 (a)	1	MA-M1 Modelling Financial Situations	MA12-2	2
14 (b)	2	MA-M1 Modelling Financial Situations	MA12-2	2-3
15	2	MA-T1 Trigonometry and Measure of Angles	MA11-3	3-4
16	2	MA-E1 Logarithms and Exponentials	MA11-5	3-4

Question	Marks	Content	Syllabus outcomes	Targeted performance bands
17	2	MA-F2 Graphing Techniques MA-E1 Logarithms and Exponentials	MA12-1 MA11-6	3-4
18 (a)	1	MA-S3 Random Variables	MA12-8	2-3
18 (b)	2	MA-S3 Random Variables	MA12-8	3-4
19 (a)	2	MA-S2 Descriptive Statistics and Bivariate Data Analysis	MA12-8 MA12-9	3-4
19 (b)	1	MA-S2 Descriptive Statistics and Bivariate Data Analysis	MA12-8 MA12-9	3
19 (c)	1	MA-S2 Descriptive Statistics and Bivariate Data Analysis	MA12-8	2-3
19 (d)	1	MA-S2 Descriptive Statistics and Bivariate Data Analysis	MA12-8 MA12-10	3-4
20 (a)	1	MA-M1 Modelling Financial Situations	MA12-4	3
20 (b)	2	MA-M1 Modelling Financial Situations	MA12-4	4-5
21 (a)	1	MA-S2 Descriptive Statistics and Bivariate Data Analysis	MA12-8	2-3
21 (b)	2	MA-S2 Descriptive Statistics and Bivariate Data Analysis	MA12-8	3-5
21 (c)	1	MA-S1 Probability and Discrete Probability Distributions	MA11-7	3-4
22 (a)	2	MA-F1 Working with Functions	MA11-2	4
22 (b)	2	MA-F1 Working with Functions	MA11-2	4
23	3	MA-C4 Integral Calculus	MA12-7	4-5
24 (a)	2	MA-C1 Introduction to Differentiation	MA11-5	4-5
24 (b)	2	MA-C4 Integral Calculus	MA12-7	4-5
25 (a)	2	MA-S1 Probability and Discrete Probability Distributions	MA11-7	3-5
25 (b)	1	MA-S1 Probability and Discrete Probability Distributions	MA11-7	3
25 (c)	2	MA-S1 Probability and Discrete Probability Distributions	MA11-7	3-4
26	3	MA-F1 Working with Functions	MA11-2	4-5
27 (a)	3	MA-S1 Probability and Discrete Probability Distributions	MA11-7	3-5
27 (b)	1	MA-S1 Probability and Discrete Probability Distributions	MA11-7	3
28 (a)	1	MA-F1 Working with Functions	MA11-2	2-3
28 (b)	2	MA-T3 Trigonometric Functions and Graphs	MA12-5	3-4
28 (c)	2	MA-T3 Trigonometric Functions and Graphs	MA12-5	3-4
29 (a)	1	MA-M1 Modelling Financial Situations	MA12-4	3
29 (b)	2	MA-M1 Modelling Financial Situations	MA12-4	4-5
29 (c)	1	MA-M1 Modelling Financial Situations	MA12-2	3
30	3	MA-E1 Logarithms and Exponentials	MA11-6	5

Question	Marks	Content	Syllabus outcomes	Targeted performance bands
31 (a)	2	MA-C3 Applications of Differentiation MA-F1 Working with Functions	MA12-3 MA11-2	4-5
31 (b)	2	MA-C3 Applications of Differentiation	MA12-3	4-5
31 (c)	2	MA-C3 Applications of Differentiation MA-F1 Working with Functions	MA12-3 MA11-2	3-4
32	3	MA-S3 Random Variables	MA12-8	5
33	3	MA-C1 Introduction to Differentiation	MA11-5	5-6
34 (a)	2	MA-T1 Trigonometry and Measure of Angles	MA11-3	4-5
34 (b)	4	MA-C3 Applications of Differentiation	MA12-6	5-6
35	5	MA-C4 Integral Calculus	MA12-7	6